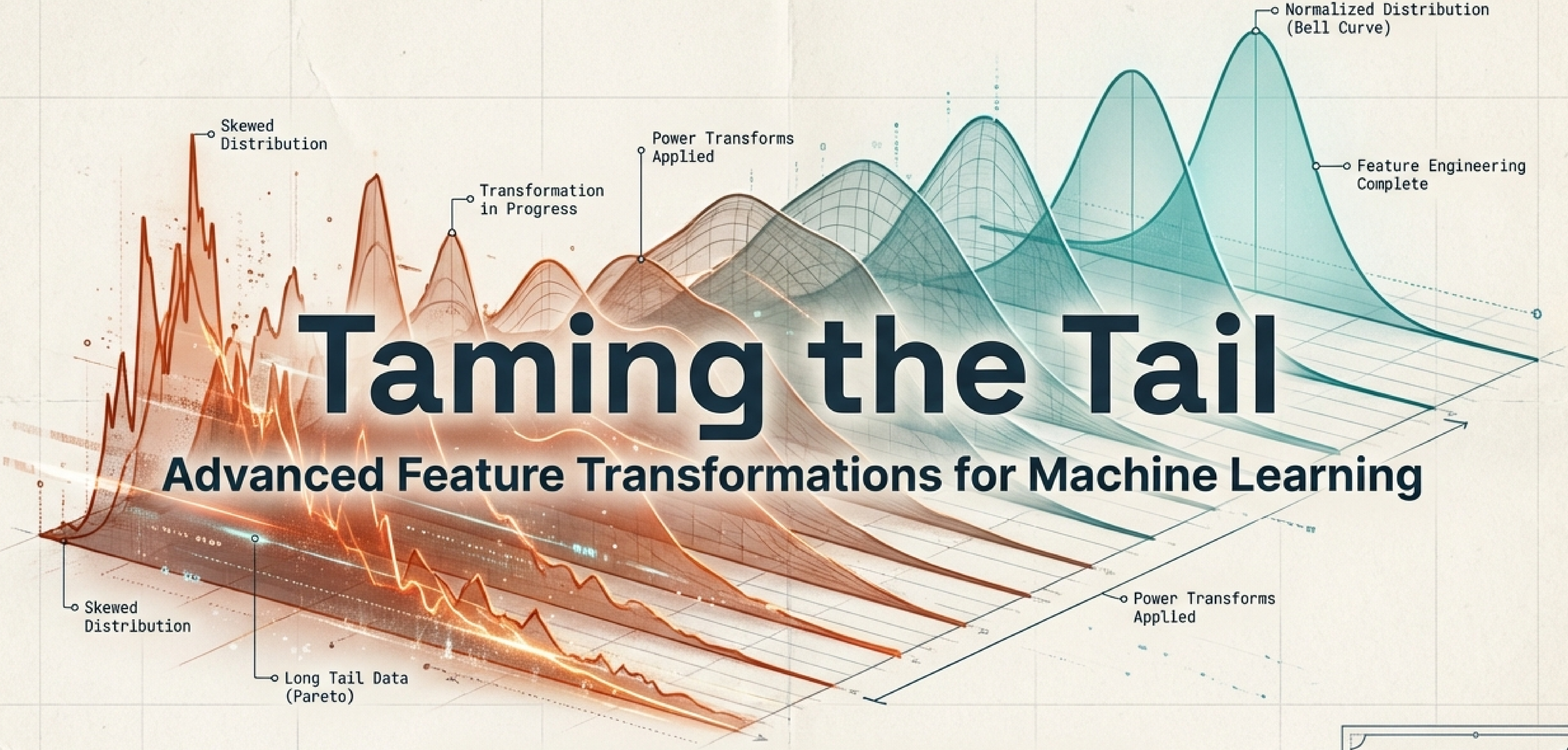
# Taming the Tail

## Advanced Feature Transformations for Machine Learning

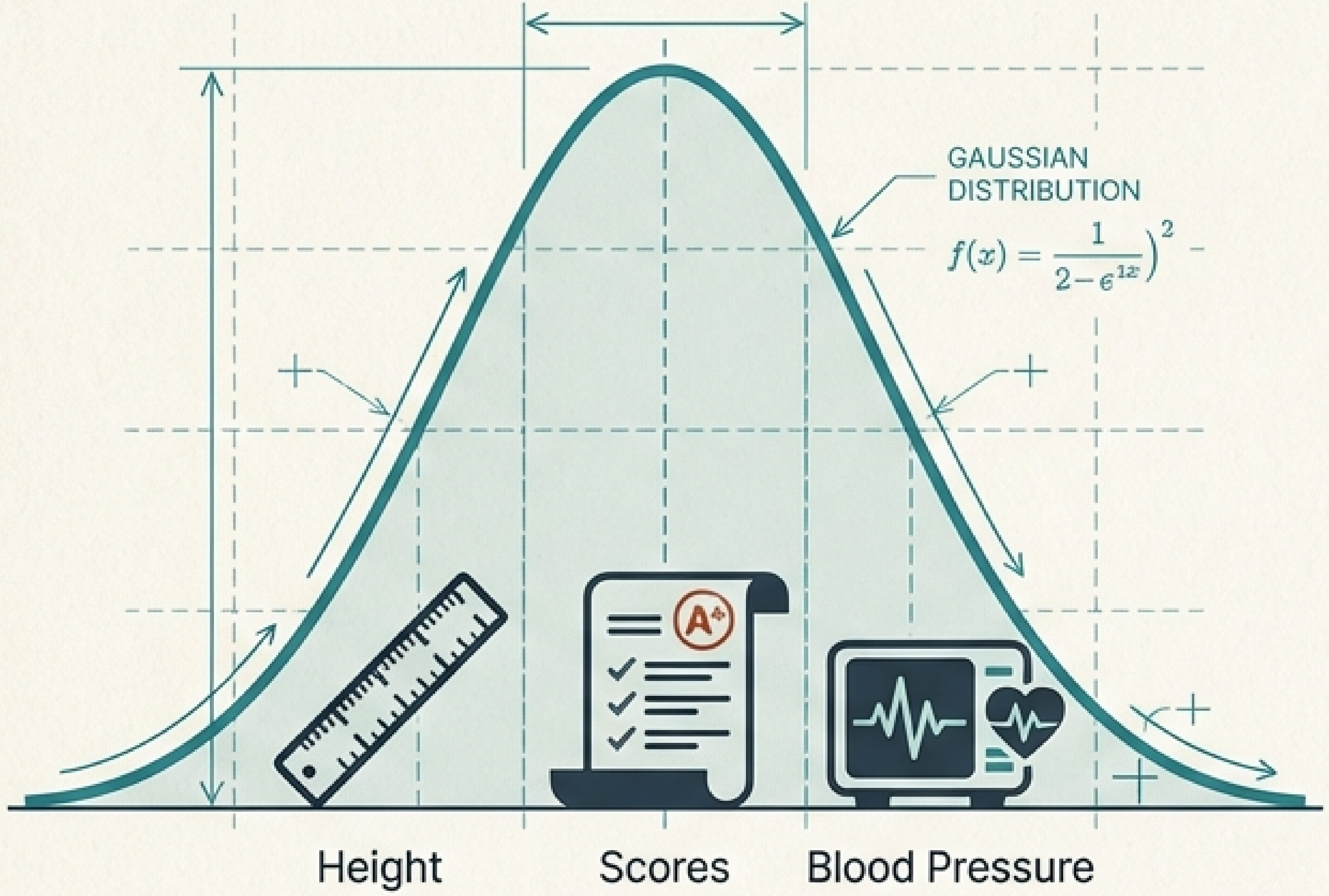



From Pareto Principles to Power Transformers
(Box-Cox & Yeo-Johnson)

Technical Reading Deck

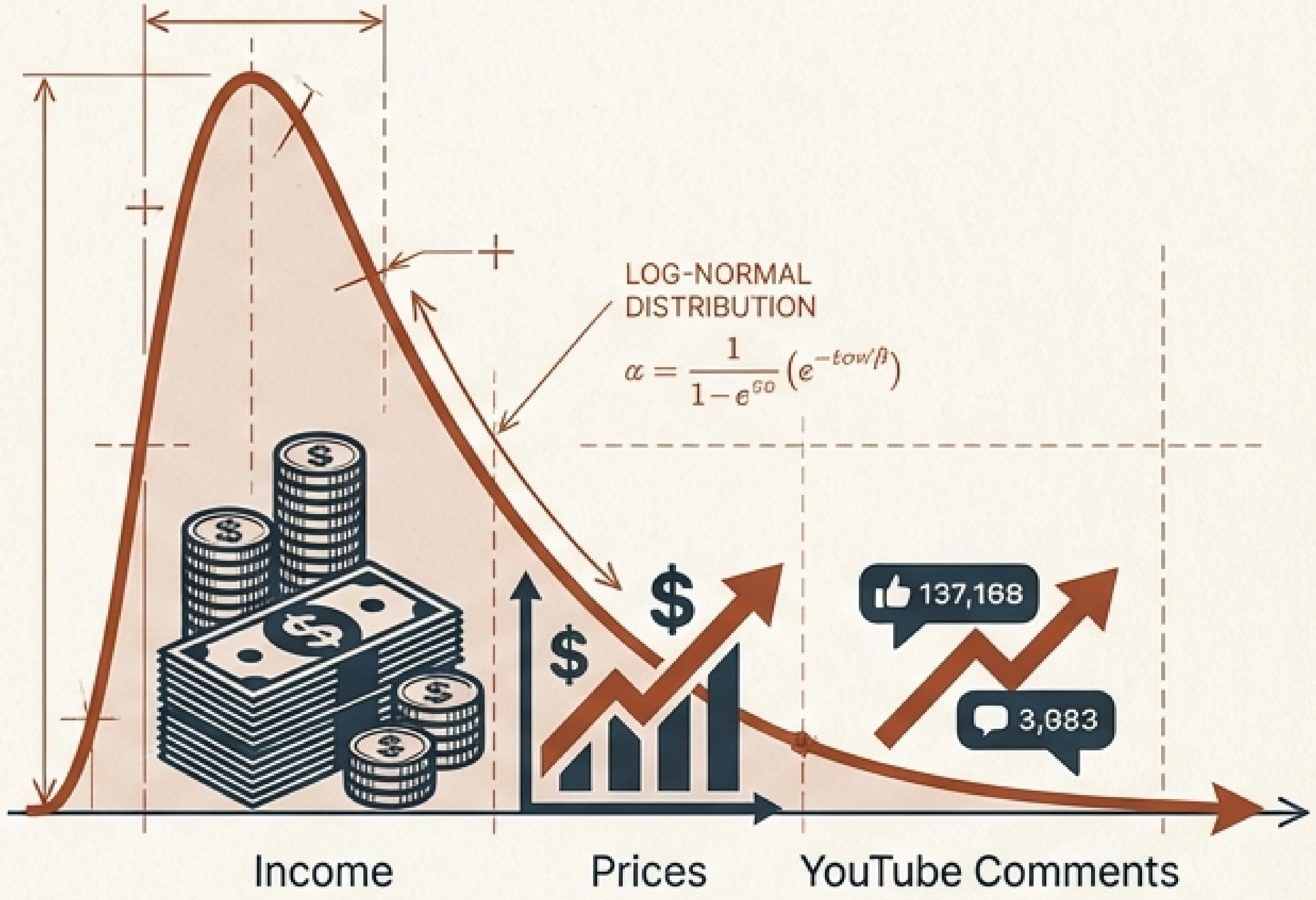
# The Myth of 'Normal' vs. The Reality of the World

## The Additive World



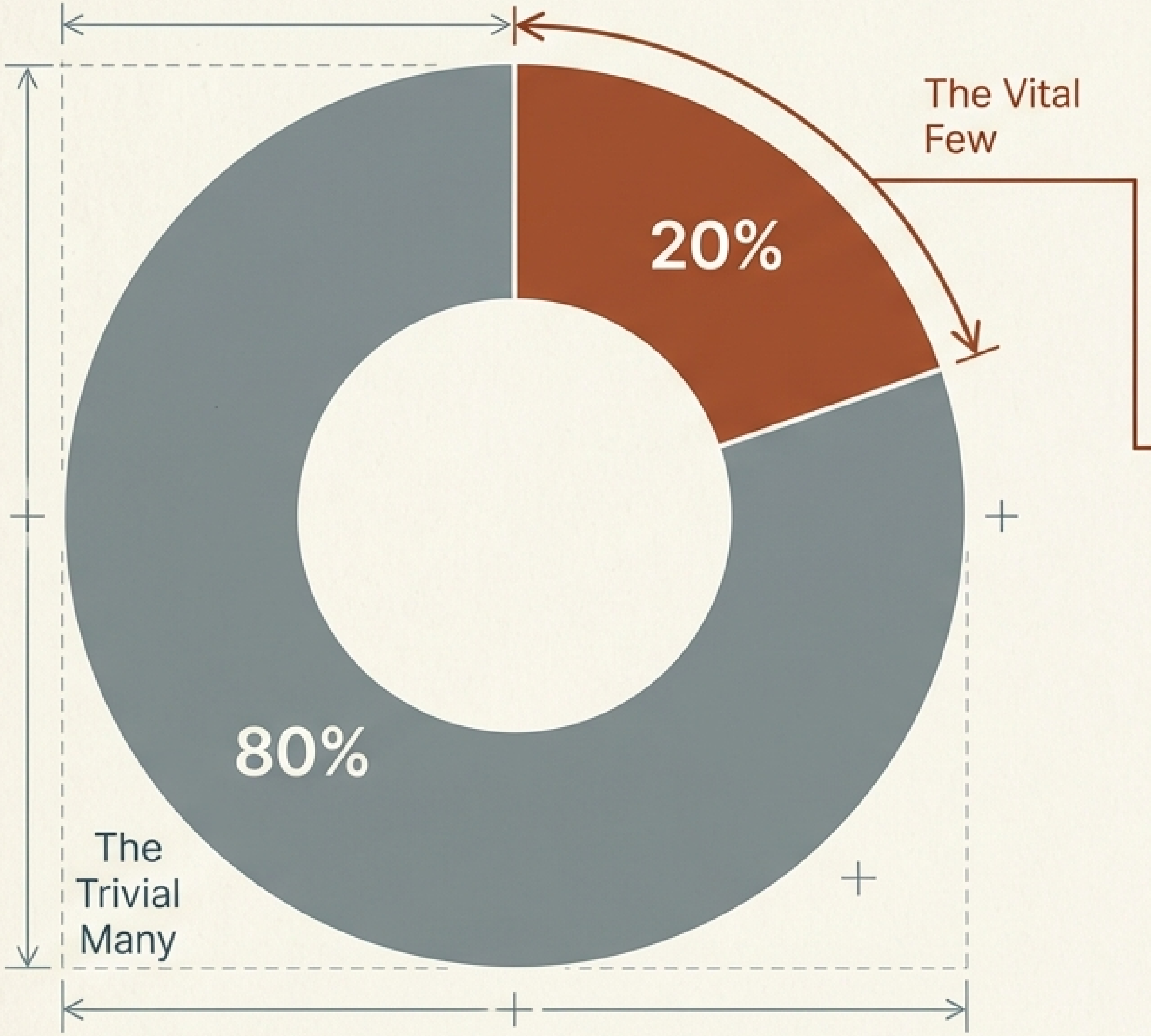

Nature adds up. Values cluster around an average.

## The Multiplicative World




Economics multiplies. Outliers define the range.

# The 80/20 Rule: Understanding the Power Law

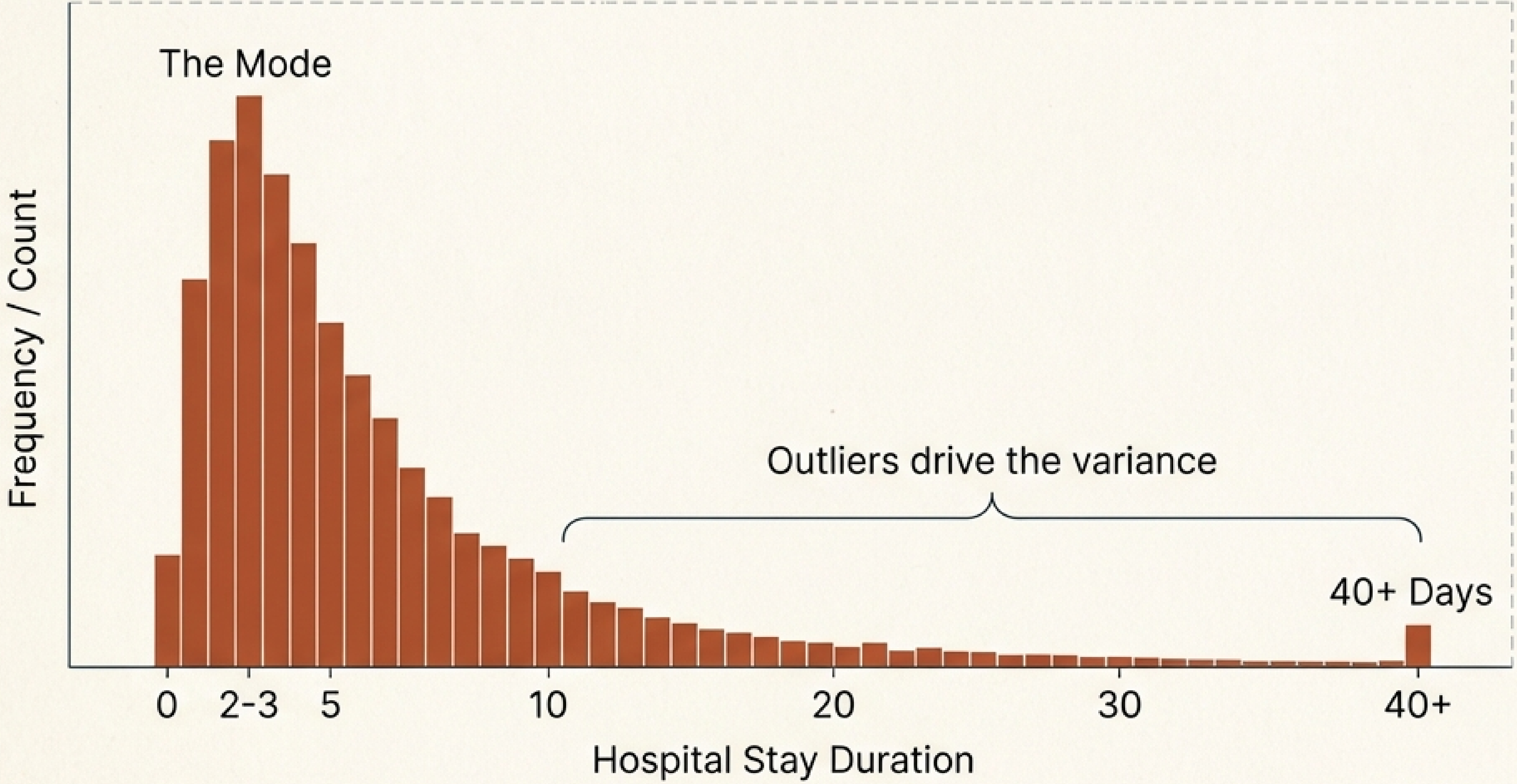



- **Wealth:** 20% of people hold 80% of global assets.
- **Sports:** 20% of players contribute to 80% of match wins.
- **Tech:** 20% of bugs cause 80% of software crashes.
- **Retail:** 20% of products drive 80% of revenue.

*Skewness isn't an anomaly.*
*It is the governing law of real-world data.*

Computational Editorial

# Log-Normal Distribution: When the Tail Wags the Dog




**Definition:**
A variable whose logarithm is normally distributed.

**Common in:**
Time-to-failure,
Income,
Duration data.

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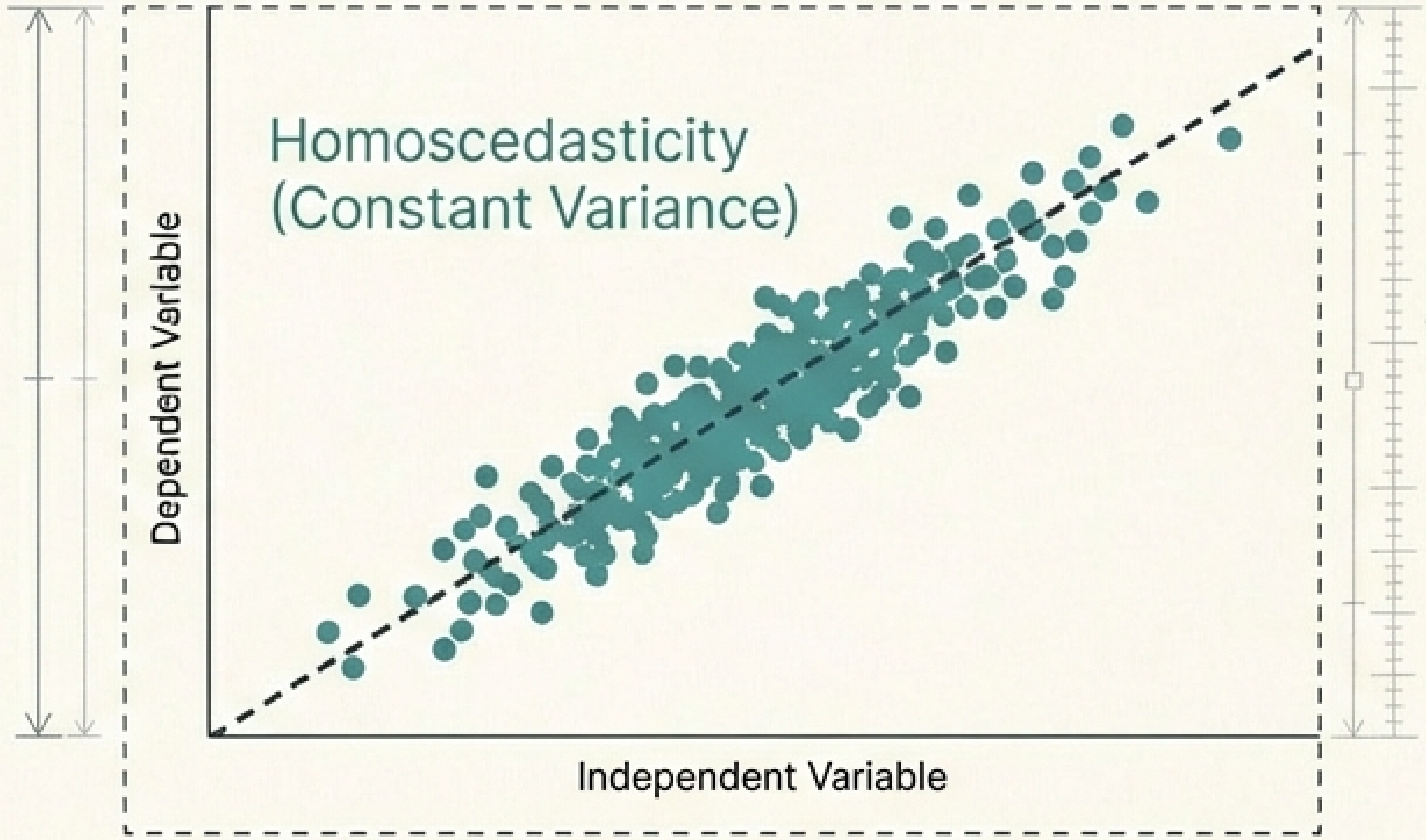
# Why Skewness Breaks Your Model

## The Problem (Bad Fit)


## The Assumption (Good Fit)




Linear algorithms assume error gradients are normally distributed. Skewness acts like a lever, pulling coefficients toward outliers and destroying predictive accuracy.

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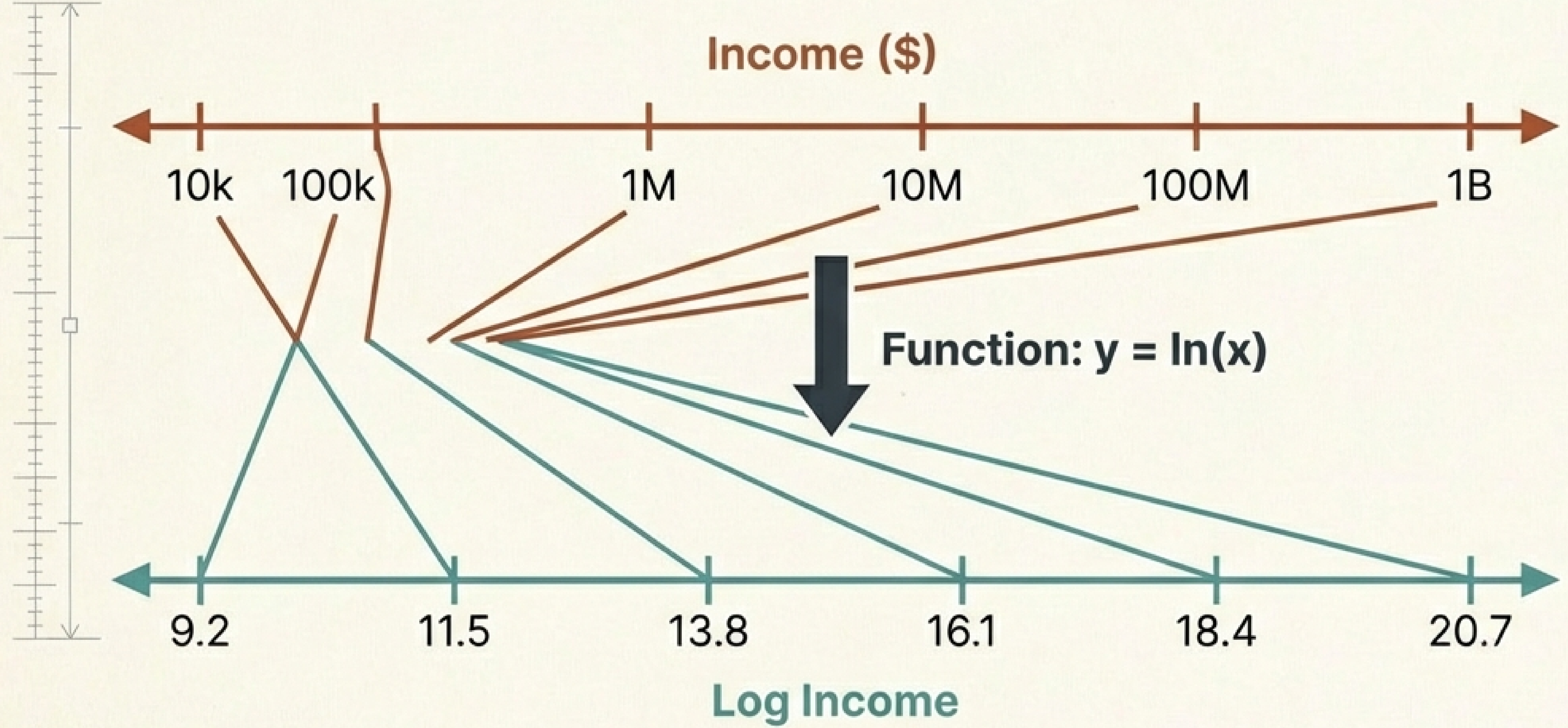
# The First Line of Defense: Log Transformation




**Logarithms** convert multiplicative relationships into additive ones. Large values shrink aggressively; small values shrink slightly.

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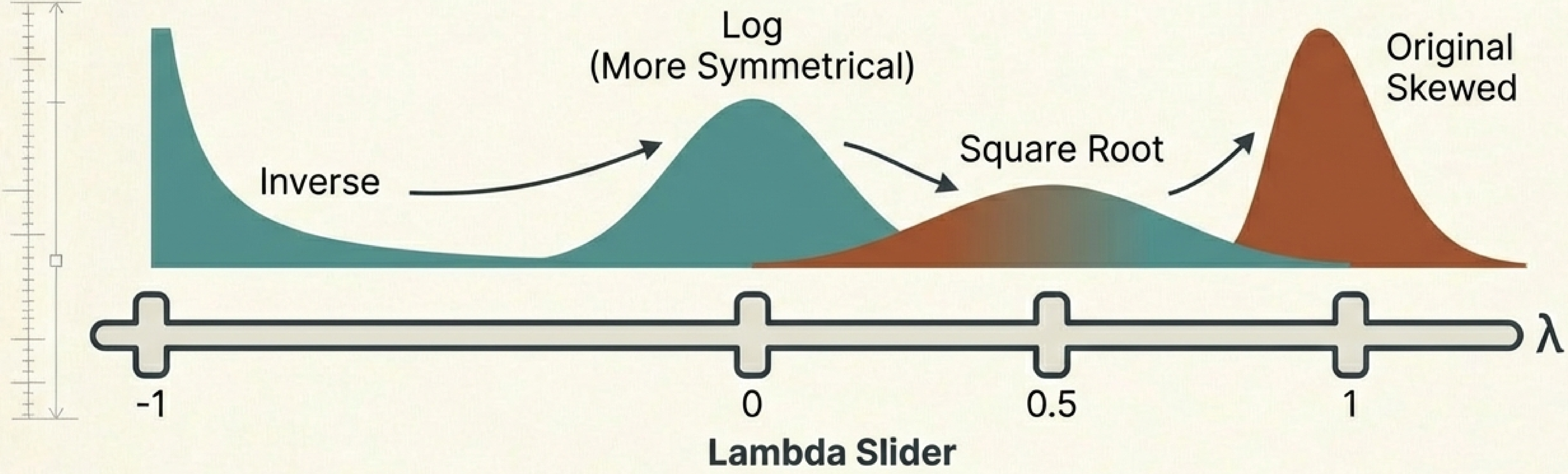
# Enter the Power Transformers

Automating the search for the perfect shape.




The algorithm tests different powers (Lambda) to minimize skewness and stabilize variance automatically.

NotebookLM

# The Box-Cox Transform

$$y(\lambda) = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(y) & \text{if } \lambda = 0 \end{cases}$$

**CRITICAL CONSTRAINT: Strictly Positive Data Only ($y > 0$).**

If your dataset contains zeros (e.g., unpaid invoices, zero defects), Box-Cox will fail unless you apply a manual shift ($y + c$).

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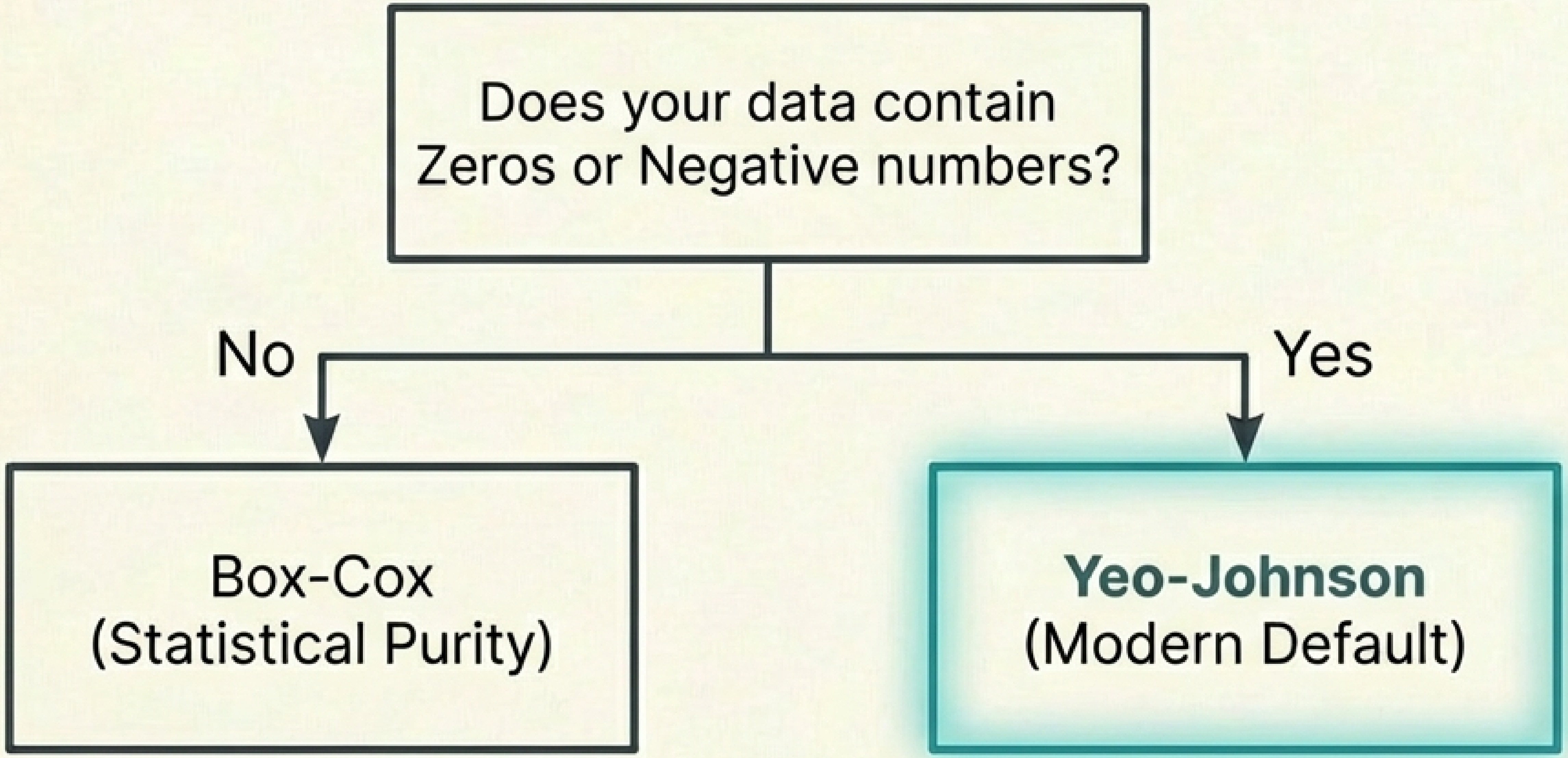
# The Yeo-Johnson Transform

The flexible, modern alternative.




A modified formula that supports real number lines ($+//0$). It is the default method in Scikit-Learn's PowerTransformer.

# The Transformer Decision Matrix

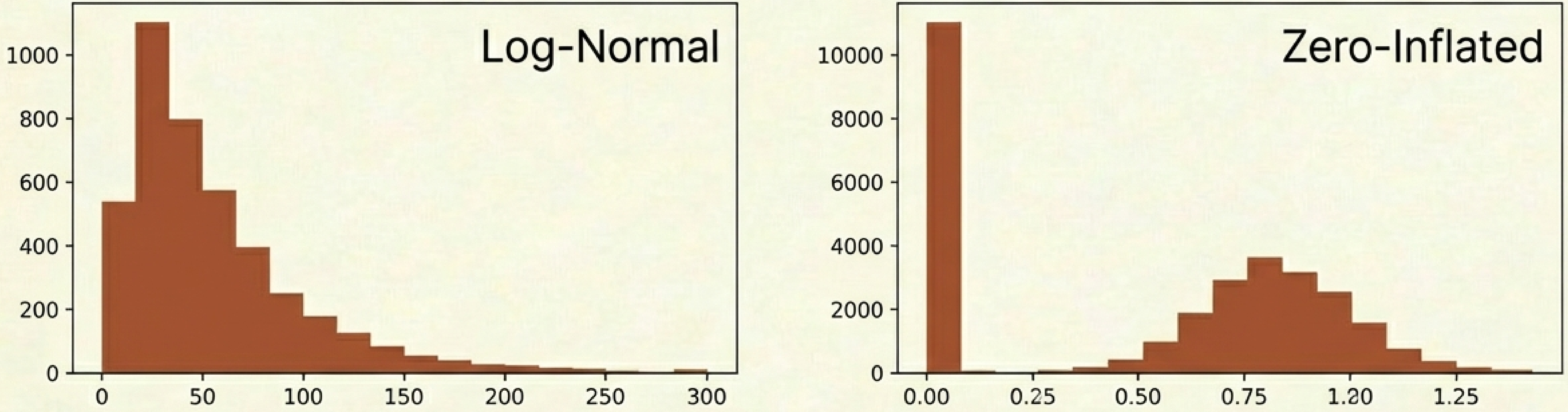
| | Log Transform | Box-Cox | Yeo-Johnson |
|---|---|---|---|
| **Complexity** | Low | High (Parametric) | High (Parametric) |
| **Parameters** | None | Lambda ($\lambda$) | Lambda ($\lambda$) |
| **Constraints** | Positive Only | Strictly Positive (>0) | All Real Numbers |
| **Best Use** | Quick Baseline | Clean Positive Data | Modern ML Pipelines |

# Case Study: Predicting Concrete Strength

## Dataset: UCI Machine Learning Repository

Feature: Age

Feature: Blast Furnace Slag




Real-world data is messy. ‘Age’ is skewed. ‘Slag’ contains zeros.
A standard Linear Regression model will struggle here.

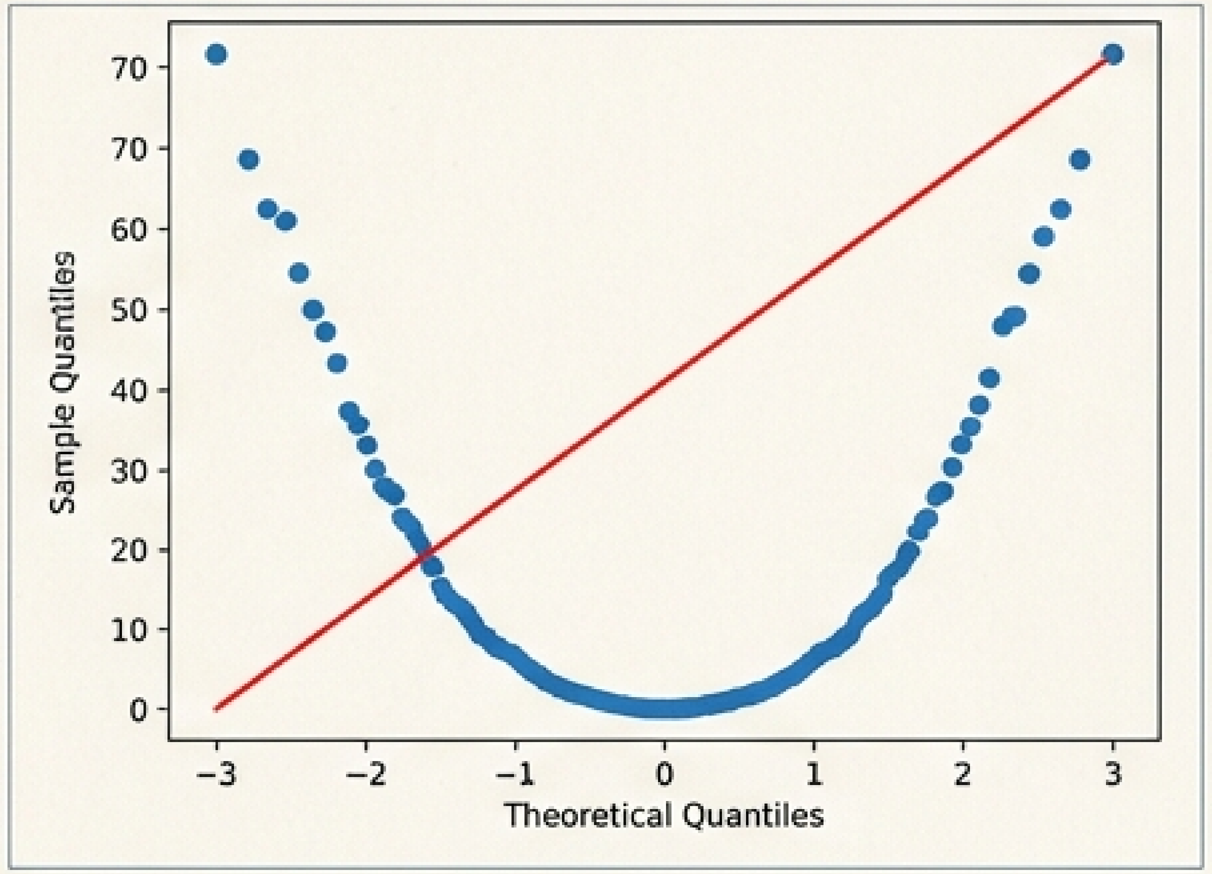
# Implementation in Scikit-Learn

```python
from sklearn.preprocessing import PowerTransformer

# Instantiate Yeo-Johnson (Default for robust handling)
pt = PowerTransformer(method='yeo-johnson')

# Fit and Transform the training data
# Automatically standardizes (zero mean, unit variance)
X_train_transformed = pt.fit_transform(X_train)

# Access the optimal Lambdas found for each column
print(f"Optimal Lambdas: {pt.lambdas_}")
```

Calculates optimal $\lambda$ for every column independently.

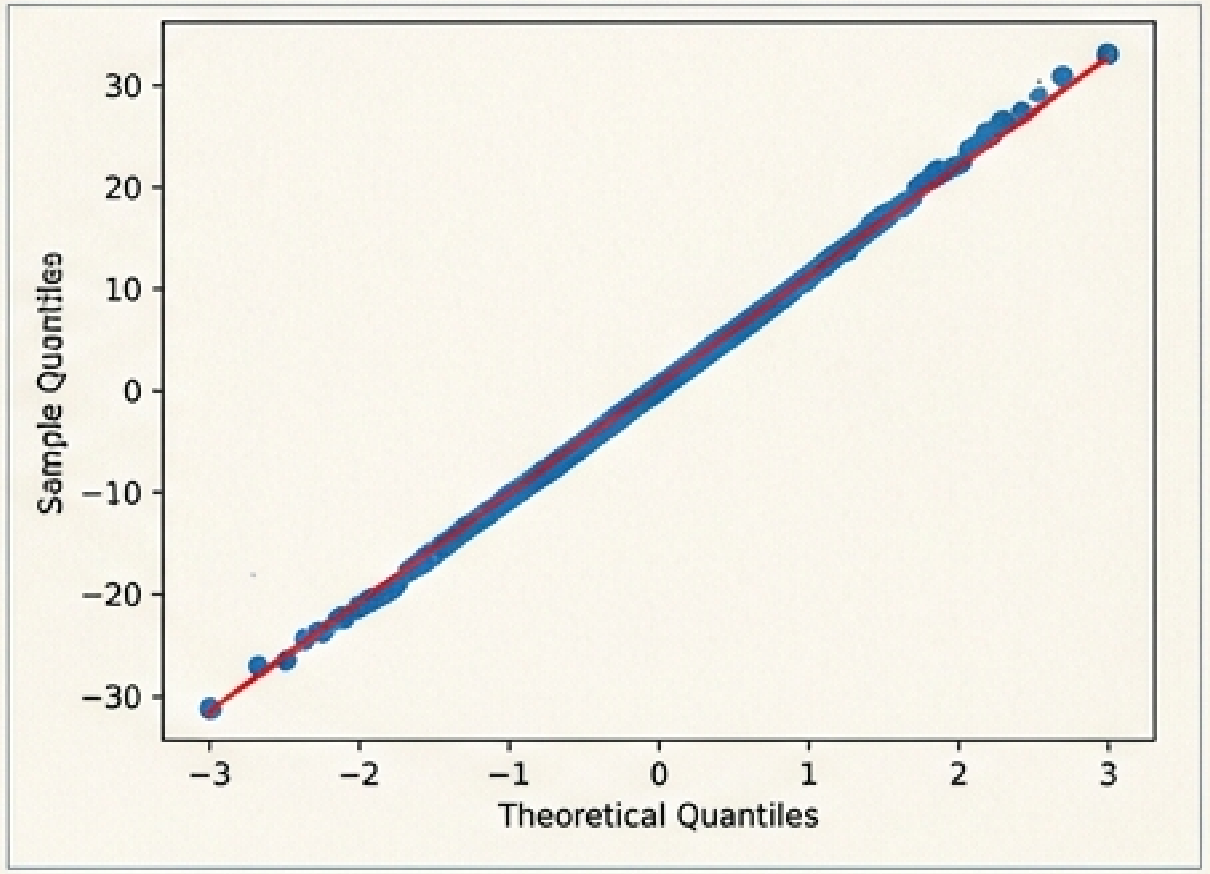
# Visual Verification: The QQ Plot

Before Transformation




Non-Normal / Skewed

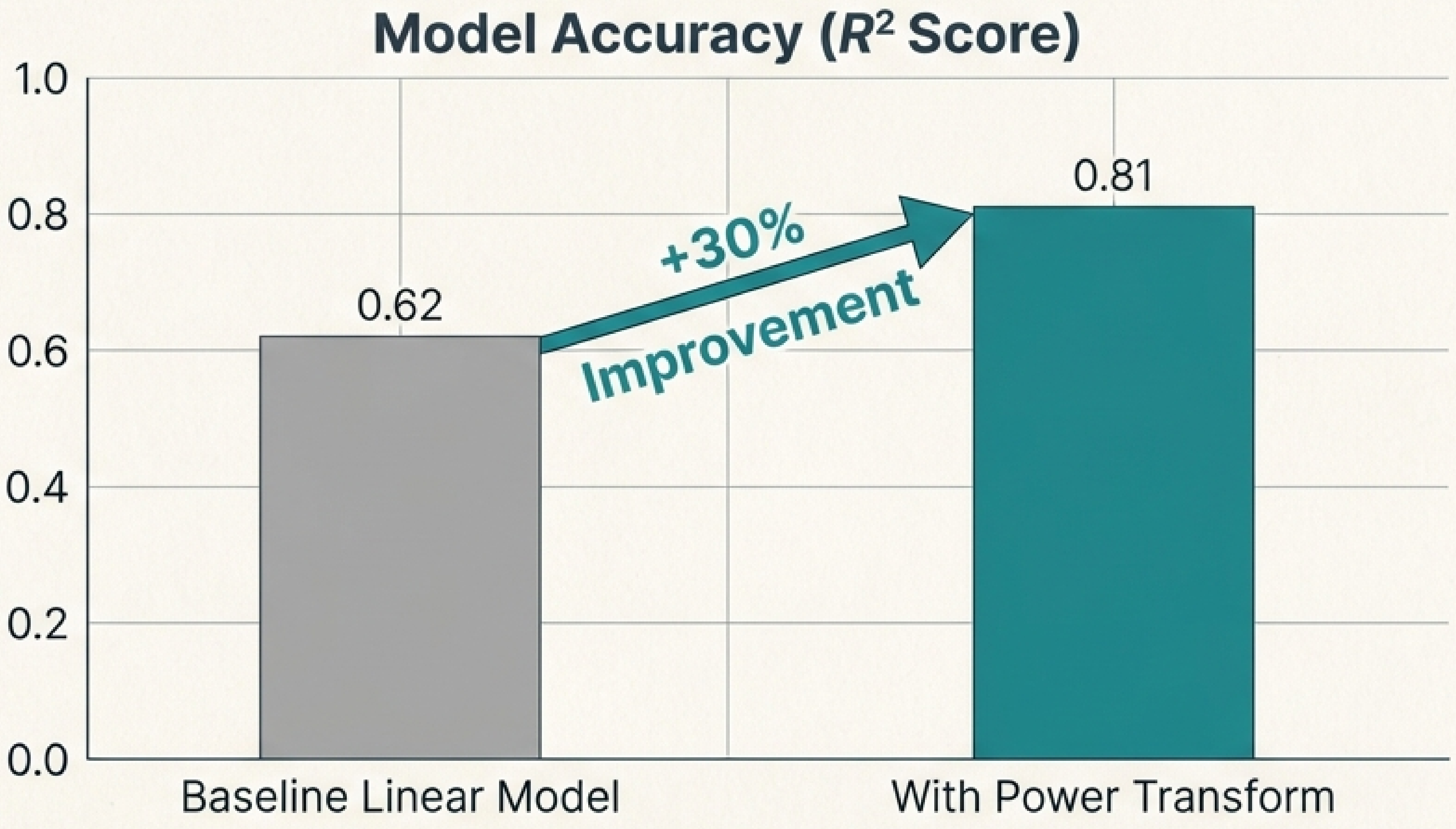
After Yeo-Johnson




Normal / Gaussian

The Probability Plot (QQ Plot) confirms that the chaotic input features have been successfully mapped to a normal distribution.

# The Bottom Line: Impact on Performance




Preprocessing with Power Transforms yielded a significant accuracy boost with zero changes to the model architecture itself.

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# Key Takeaways

1. **Diagnosis**: Real-world data (Wealth, Errors, Time) follows **Pareto/Log-Normal** distributions, not Bell Curves.
2. **Conflict**: Skewed data drags down the accuracy of linear ML models.
3. **Treatment**: Use **Box-Cox** for strictly positive data. Use **Yeo-Johnson** for data with zeros or negatives.
4. **Result**: PowerTransFormer finds the optimal shape (λ), stabilizing variance and boosting $R^2$ scores.